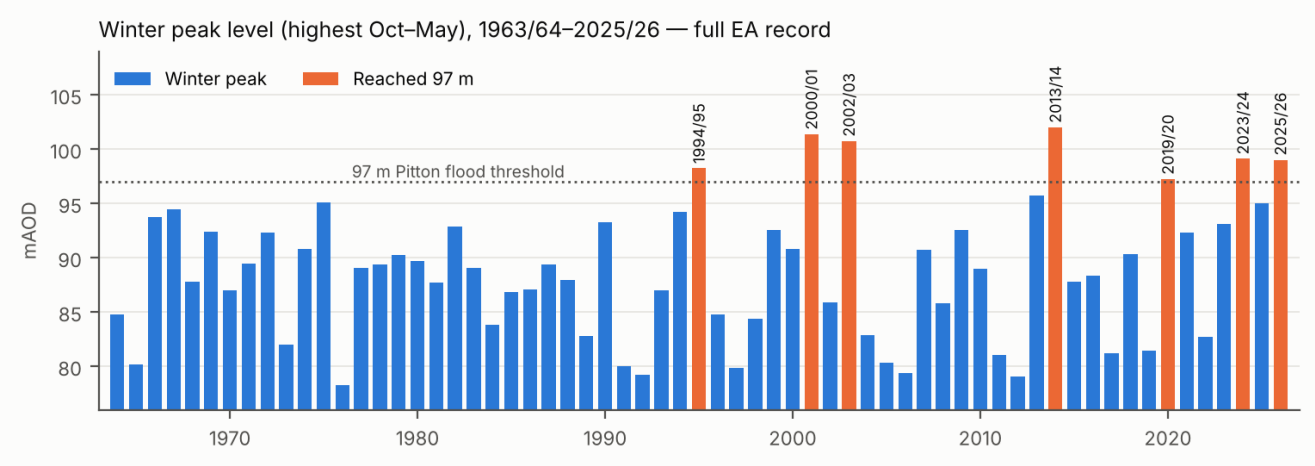
Lopcombe Corner Borehole — Summary of Findings

EA station E14150 (Chalk, near Winterslow) · levels: full EA record 1963–2026 (manual dips and daily logger) · rainfall: Met Office HadUK-Grid 1 km for the borehole's cell · 20 Oct 2026

Record: 102.01 mAOD on 13 Feb 2014. **Floods:** 7 of 63 winters reached the 97 m Pitton groundwater-flood threshold — 1994/95, 2000/01, 2002/03, 2013/14, 2019/20, 2023/24, 2025/26 — five of them since 2000 and four since 2013. **Now:** 78.25 m on 2026-10-20, a normal autumn level (the well drains back to about 78.5 ± 0.4 m every autumn).



What drives the winter peak

- **Winter rainfall, mainly.** October–March rainfall explains most of the peak (r = 0.83 over 62 winters): about **+3.9 m of peak per extra 100 mm**. Every flood winter with rainfall data had at least about 670 mm October–March; of the seven winters above that, six flooded (not 2022/23).
- **The starting level matters too.** Adding the 1 October level lifts the share explained from 0.69 to 0.79: each extra metre at the start adds about **4.5 m** to the peak ($p < 0.001$). A high autumn start is worth roughly 50 mm of extra winter rain.
- **Timing within the winter** explains much of what is left — intense February rain (2019/20, 1989/90) or an early-January spike followed by a dry spell (2022/23).

Cycles and trends

- **No regular cycle.** Neither rainfall nor peak level shows a significant periodicity (strongest periods 3.3 and 6.7 years, $p \approx 0.4$–0.5); wet winters come in short, irregular clusters.
- **Winter rainfall has crept up** by about 1.4 mm a year since 1963 (≈ +87 mm), but this just misses statistical significance ($p = 0.06$). Winter peak levels show no trend ($p = 0.70$), even though floods have clustered since 2000.

Early warning: chance of reaching 97 m later in the winter

Level at end of …	Oct	Nov	Dec	Jan	Feb
50:50 chance at about	not useful	88–89 m	92–93 m	93–95.5 m	~96 m
Lowest level that still went on to flood	78.2 m	78.5 m	79.6 m	89.6 m	93.7 m

- At the end of October the chance is about the long-run 1 in 10 whatever the level. From November the level is a strong guide. Every flood winter was at least 89.6 m by the end of January, and none first reached 97 m after the end of February. Percentages carry about ±10–20 points of uncertainty (only seven flood winters).

Data notes

HadUK-Grid rainfall is an estimate interpolated from surrounding gauges, not a reading at the borehole; it agrees closely with Broughton (E13580) and has no gaps. On 20 Oct 2026 the full EA level record showed that several logger-era winter peaks used earlier had been read too low (e.g. 2023/24 is 99.11 m, not 97.54 m); all current reports use the corrected record. Details, methods and scripts: docs/reports and docs/notes in the Lopcombe_Corner_GWL repository.